Step 1: Start Docker Environment

docker-compose up -d

```
C:\Users\HP>D:

D:\Module5_Docker_Environment>docker-compose up -d
[+] up 42/42
 ✔Image postgres:16          Pulled                                                                209.1ss
 ✔Image mysql:8              Pulled                                                                243.5ss
 ✔Image ollama/ollama:latest Pulled                                                                1227.9s
 ✔Image minio/mc:latest      Pulled                                                                29.4sss
#1 [internal] load local bake definitions
#1 reading from stdin 575B done
#1 DONE 0.0s
```

Step 2: Verify Containers

docker ps

```
D:\Module5_Docker_Environment>docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED         STATUS                   PORTS
                    NAMES
70666a6715d0   trinodb/trino:latest           "/bin/sh -c 'cp -r /…"   4 minutes ago   Up 4 minutes (healthy)   8080/tcp
                    module5_docker_environment-trino-worker-1
13b4af333e2b   module5_docker_environment-ingestor   "/bin/sh -c 'echo 'W…"   4 minutes ago   Up 4 minutes
                    ingestor
1431fa8a4b2c   trinodb/trino:latest           "/usr/lib/trino/bin/…"   4 minutes ago   Up 4 minutes (healthy)   0.0.0.0:13579->8080/tcp, [::]:13579->
8080/tcp            trino-coordinator
2756e0bb54d2   minio/mc:latest                "/bin/sh -c 'sleep 5…"   4 minutes ago   Up 4 minutes
                    module5_docker_environment-mc-1
97a4969ff4f7   postgres:16                    "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes             0.0.0.0:5434->5432/tcp, [::]:5434->54
32/tcp              pg_catalog
91be03ceff1a   postgres:16                    "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes             0.0.0.0:5433->5432/tcp, [::]:5433->54
32/tcp              pg_source
911994a36153   mysql:8                        "docker-entrypoint.s…"   4 minutes ago   Up 4 minutes             0.0.0.0:3307->3306/tcp, [::]:3307->33
06/tcp              mysql_source
35df525d863a   ollama/ollama:latest           "/bin/ollama serve"      4 minutes ago   Up 4 minutes             0.0.0.0:11434->11434/tcp, [::]:11434-
>11434/tcp          ollama
60bc035be878   minio/minio:latest             "/usr/bin/docker-ent…"   4 minutes ago   Up 4 minutes             0.0.0.0:9000-9001->9000-9001/tcp, [::
]:9000-9001->9000-9001/tcp   minio
```

Step 3: Run Ollama Model

docker exec -it ollama ollama run qwen2.5:3b

```
D:\Module5_Docker_Environment>docker exec -it ollama ollama run qwen2.5:3b
```

```
>>> Generate ONLY valid Trino SQL query.
... No explanation.
...
... Table: iceberg.sales_schema.sales
... Columns: order_id, product, category, price, quantity, order_date
...
... Task: Calculate total revenue using SUM(price * quantity)
```sql
SELECT SUM(price * quantity) AS total_revenue
FROM iceberg.sales_schema.sales;
```
```

Step 4: Exit Ollama

```
>>> /bye
```

Step 5: Navigate to Project Folder

D:

cd Module5_Docker_Environment\Module5_Docker_Environment

```
(base) D:\>cd D:\Module5_Docker_Environment
```

Step 6: Run Python Script

```
(base) D:\Module5_Docker_Environment\Module5_Docker_Environment>python llm_trino_query.py
Enter your question: total revenue
```

Step 7: Provide Natural Language Input

```
Enter your question: total revenue
```

Step 8: LLM Generates SQL

```
Generated SQL:
 SELECT SUM(price * quantity) AS total_revenue
FROM iceberg.sales_schema.sales;
```

Step 9: Execute Query & Display Result

```
(base) D:\Module5_Docker_Environment\Module5_Docker_Environment>python llm_trino_query.py
Enter your question: total revenue

Generated SQL:
 SELECT SUM(price * quantity) AS total_revenue
FROM iceberg.sales_schema.sales;

Query Result:
 "110000.0"
```